ZIHAN (MICHAEL) WANG

415-535-9521 | zihan16@illinois.edu | linkedin.com/in/zihan32/

EDUCATION

University of Illinois Urbana-Champaign - Champaign, IL Expected May 2027
Bachelor of Science in Mechanical Engineering
Relevant Coursework: Mechanical Design, Controls, Fluid Mechanics, Heat Transfer, Thermodynamics, Materials Engineering

TECHNICAL SKILLS

CAD/Design: SolidWorks, AutoCAD, Fusion 360, Onshape, Rhino 7/8, FEA
Programming/Analysis: Python, MATLAB/Simulink, Arduino, Java, Excel, G-code
Manufacturing: CNC Machining, 3D Printing, Laser Cutting, MIG Welding, Soldering, Injection Molding
Engineering: Engineering drawings, prototyping, experimental testing, data analysis, design iteration

ENGINEERING EXPERIENCE

Proton Products International Ltd. - R&D Engineering Intern Jun 2025 - Aug 2025
Chengdu, China
- Supported new-product development for non-contact diameter measurement systems through mechanical design, prototype iteration, and engineering testing.
- Created and revised 3D CAD models and engineering drawings for prototype components based on design requirements, test observations, and R&D feedback.
- Assisted with prototype assembly and functional testing, troubleshooting fit, alignment, manufacturability, and performance issues during design iteration.
- Participated in design reviews and incorporated prototype test observations into mechanical design updates and engineering documentation.

Proton Products International Ltd. - Mechanical Engineering Intern Jun 2024 - Aug 2024
Chengdu, China
- Assembled and calibrated 10 non-contact laser diameter gauges and verified measurement performance prior to customer delivery.
- Conducted accuracy testing using proprietary measurement software, contributing to a 100% customer acceptance rate for completed units.
- Supported development of a single-axis laser scanning diameter gauge by producing 3D CAD models, AutoCAD drawings, and prototype design updates.

Meyer Lab, UC San Diego - Research Apprentice Jul 2023 - Aug 2023
- Developed CAD models of bio-inspired structures to support investigation of geometry, structural behavior, and mechanics-based design concepts.
- Assisted with research integrating theoretical, experimental, and computational approaches in mechanics and materials.

ENGINEERING PROJECTS

Project TerraTrek - Legged Robot Mechanism (ME 370) Fall 2026 - Present
- Developing a compact, manually operated legged mechanism to advance a stair-profile rack that simulates stepped and sloped agricultural terrain within 24 x 24 x 24 cm and $60 design constraints.
- Applying mechanism synthesis, concept evaluation, and iterative low- and mid-fidelity prototyping to improve stable motion and prevent back-sliding under a single-crank input.
- Evaluating linkage concepts through structured concept selection, prototype testing, and design reviews before final mechanism demonstration.

Mechanical Properties & Experimental Testing Fall 2026 - Present
- Process tensile-test data for engineering alloys and generate stress-strain curves to determine elastic modulus, 0.2% offset yield strength, ultimate tensile strength, and fracture strain.
- Analyze hardness and tensile behavior across engineering alloys using statistical methods to compare material properties and experimental variability.

Dynamic Systems & Controls Analysis Summer 2026
- Modeled mechanical dynamic systems using transfer functions and state-space representations in MATLAB/Simulink and evaluated transient response under multiple inputs.
- Analyzed stability, frequency response, Bode behavior, and steady-state system response to support controller and system-level design decisions.

ENGINEERING TEAMS

Illini Electric Motorsports - Mechanical Team Member Oct 2022 - Present
- Contribute to design, manufacturing, and vehicle-integration activities for Formula SAE systems using CAD, FEA, and design-for-manufacturability principles.
- Collaborate with multidisciplinary subteams during component fabrication, vehicle assembly, testing, and iterative design improvements.

Illini Solar Car - Mechanical Team Member Oct 2022 - Present
- Contribute to development of a student-built solar-electric race vehicle, including carbon-fiber chassis and aluminum suspension design and fabrication activities.
- Apply FEA, physical testing, aerodynamic model iteration, and vehicle-integration methods to improve structural integrity and vehicle efficiency.